

Integer Factorization: Another perspective

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ABSTRACT

Integer factorization is a fundamental problem in algorithmic number theory and computer science. This problem is considered as a one way or trapdoor function in cryptography, mainly in the Rivest, Shamir and Adleman (RSA) cryptosystem where it constitutes the foundation of the key exchange protocol and digital signature. To date, from elementary trial division to sophisticated methods like the General Number Field Sieve, no known algorithm can break the problem in polynomial time, while it is proved that Shor's algorithm could on a quantum computer. That said, these methods are limited in term of efficiency and their success strongly linked to computational resources, particularly with the Shor's algorithm using quantum circuits that entirely depends on a quantum computer. In this paper, we start by recalling slightly some factorization facts and then approach the problem under completely different angles. Firstly, we take the problem from the ring $(\mathbb{Z}, +, \cdot)$ to the Lebesgue space $\mathcal{L}^1(X)$ where X can be $\mathbb{Q}$ or any given interval setting. From this first perspective, integer factorization becomes equivalent to finding the perimeter of a rectangle whose area is known. In this case, it is equivalent to either finding bounds of integrals or finding primitives for some given bounds. Secondly, we take the problem from the ring $(\mathbb{Z}, +, \cdot)$ to the ring of matrices $(M_2(\mathbb{Z}), +, \cdot)$ and show that this problem is equivalent to matrix decomposition, and therefore present some possible computing algorithms, particularly using Gröbner basis and through matrix diagonalization. Finally, we address the problem depending on algebraic forms of factors and show that this problem is equivalent to finding small roots of a bivariate polynomial through Coppersmith's method.

The aim of this study is to propose innovative methodological approaches to reformulate this problem, thereby offering new perspectives.

1 INTRODUCTION

Factoring integers is one of the most important problems in cryptography and number theory. In 1801, in "Disquisitiones Arithmeticae" [16], Gauss identified integer factorization and the primality testing as the two fundamental problems of arithmetic. Indeed, he went on to say: "The dignity of science itself seems to require that all possible means should be explored to solve a so elegant and famous problem." (Translation of the original texts). These two main problems are at the heart of cryptographic protocols, to ensure the confidentiality, authenticity, integrity and non-repudiation of information. This is the case of the Diffie-Hellman Elliptic Curve (ECDH) protocol implemented in TLS, for example, where the primality of the cardinal of the associated group is an important security feature to avoid attacks like MOV; the Elliptic Curve Digital Signature (ECDSA) protocol, or the RSA cryptosystem, whose security essentially relies on integer factorization.

Over the 20th century, a number of factoring algorithms were proposed, the most important of which are the Quadratic Sieve (QS) proposed by Carl Pomerance [17] in 1985 and the General Number Field Sieve (GNFS) proposed by the same author 9 years later [18] in 1994, based on the Factorisation method proposed by Pierre de Fermat a century earlier [33], [55], in addition the Elliptic Curve Method (ECM) proposed by Hendrik Lenstra [40] in 1987. These methods have also seen significant improvements during this century (see [7], [62], [20], [3], [64], [6], [49]). Since the beginning of the 21st century, several other improvements to these methods have been proposed, particularly for the ECM method, both in terms of parametrizations and addition formulas for different coordinate systems ([26], [41], [11], [14], [12], [13], [64], [59], [30]).

In practice however, these algorithms still present feasibility limits for fairly large parameters. This is the case, for example, of the record set in 2019 by Emmanuel Thomé, Paul Zimmermann, Fabrice Boudot, Pierrick Gaudry, Aurèle Guillemin and Nadia Heninger on a 795-bit RSA module (RSA-240) using an optimized implementation of the state-of-the-art method, the number-field sieve method in the CADO-FS software [71] with a computation time of around 1000 physical core-years on Intel Xeon Gold 6130 CPUs [61], or even the latest feat on an 829-bit RSA module (RSA-250) achieved in February 2020 by the same authors using CADO-NFS with a computation time of 2,700 core-years on Intel Xeon Gold 6130 CPUs, distributed across the computing resources of the INRIA and CNRS laboratories and their partner institutions; the EXPLOR computing center of the Université de Lorraine in Nancy (France); the PRACE research infrastructure using resources at the Juelich supercomputing center in Germany, as well as Cisco equipment at UCSD.

All these requirements in terms of hardware resources and very high computing power demonstrate how important it is to explore new directions in order to propose new approaches that could lead to efficiently solving such an important problem in cryptology and number theory.

In this paper we address the problem in another direction with a whole new point of view. That is, primitives and integral calculus on one side, matrix decomposition with application of Gröbner basis on the other side and finally polynomial small roots computation through Coppersmith method.

2 BACKGROUND

Factorization methods can be divided into 2 main categories: special purpose factorization methods and general purpose factorization methods.

We hereby consider n to be a semiprime of size b bits, p and q factors of n with $\min\{p, q\} = p$.

2.1 Special purpose factorization methods

Special purpose factorization methods running time depends on the properties of the number to be factored, mainly its factors such as the smallest prime factor. Among these algorithms, we have:

(1) Pollard's rho algorithm

The Pollard's rho algorithm is an algorithm for integer factorization invented by John Pollard in 1975. It takes as its input n the integer to be factored and considers the polynomial $g(x) = (x^2-1) \bmod n$ or more commonly $g(x) = (x^2+1) \bmod n$ and outputs a non trivial factor of n , or it fails. This method has an expected running time proportional to the square root of the smallest prime factor of the composite number being factorized, that's to say $O(\sqrt{p})$.

(2) Fermat's factorization Method

The Fermat's Factorization Method factors an odd integer n by expressing it as the difference of two squares:
 $n = a^2 - b^2 = (a-b)(a+b)$.

Its time complexity in the worse case is $O(\sqrt[4]{n})$.

(3) Euler's factorization Method

The Euler's Factorization Method is based on representing a number n as the sum of two squares in two different ways, leading to its factorization. Specifically, it uses the identity $n = a^2 + b^2 = c^2 + d^2$.

Its time complexity in the worst case is $O(\sqrt{p})$.

(4) Lenstra's elliptic curve method

The Elliptic Curve factorization Method (ECM) that was introduced by H. Lenstra [40] and conjectured to be in $O\left(\exp^{(\sqrt{2}+O(1))\sqrt{\log p \log \log p}}\right)$ (where p is the smallest factor of the composite module n) is an improvement of the pollard $p-1$ method (see [45]) where the multiplicative group $\mathbb{Z}/n\mathbb{Z}$ is replaced by $E(\mathbb{Z}/n\mathbb{Z})$, an elliptic curve defined over the residue ring $\mathbb{Z}/n\mathbb{Z}$, where n is the number to factor. From short Weierstrass to the use of Montgomery curves [62]. Since the introduction of Edwards Curve [41] and their use in the ECM [26] thanks to their large torsion subgroups by Mazur Theorem [9], Edwards Curves have replaced Montgomery curves in latest implementation ECM-MPFQ [30] over the GMP-ECM [59]. Results in [3, 11–14, 20, 23, 64], show significant improvements not only at the scalar multiplication level with fewer number of Inversions, Multiplications and Squarings but also at parametrization level.

An elliptic curve over a field $\mathbb{K}$ is an algebraic curve given by a non singular equation of the form

$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, $a_i \in \mathbb{K}$ called generalized weierstrass equation form. In the projective space $\mathbb{P}^2(\mathbb{K}) = \mathbb{K}^3 \setminus \{(0, 0, 0)\}$ through the map $(x, y) \mapsto (X/Z, Y/Z)$, the

equation becomes

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

More generally, there exist isomorphisms that change the

generalized form to reduced forms dependently on the characteristic of the field in which the curve is defined. For example

$$\begin{cases} y^2 = x^3 + ax + b & \text{if } \text{char}(\mathbb{K}) > 3 \\ y^2 = x^3 + ax^2 + bx + c & \text{if } \text{char}(\mathbb{K}) = 3 \\ y^2 + cy = x^3 + ax + b & \text{if } \text{char}(\mathbb{K}) = 2 \\ y^2 + xy = x^3 + ax^2 + b & \text{if } \text{char}(\mathbb{K}) = 2 \end{cases} \quad (1)$$

$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, a_i \in \mathbb{F}_p\} \cup \{O\}$, has an abelian group structure with the additive law where $O = (0 : 1 : 0)$ denotes the point at infinity, corresponding to the case $Z = 0$ in the projective space.

Considering a field with characteristic $p > 3$ and two points $P = (x_p, y_p), Q = (x_q, y_q) \in E_{a,b}(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 / y^2 = x^3 + ax + b, a, b \in \mathbb{F}_p\}$, using the chord and tangent rule with the theorem of Bezout [29], the addition law on the curve is defined by

$$\text{if } P \neq Q : \begin{cases} x_{P+Q} = \left(\frac{y_P - y_Q}{x_P - x_Q} \right)^2 - x_P - x_Q \\ y_{P+Q} = -y_P + \frac{y_P - y_Q}{x_P - x_Q} (x_P - x_{P+Q}) \end{cases};$$

$$\text{if } P = Q : \begin{cases} x_{P+P} = x_{2P} = \left(\frac{3x_P^2 + a}{2y_P} \right)^2 - 2x_P \\ y_{P+P} = y_{2P} = -y_P + \frac{3x_P^2 + a}{2y_P} (x_P - x_{2P}) \end{cases}$$

Setting $\alpha = \frac{y_P - y_Q}{x_P - x_Q}$ or $\alpha = \frac{dy}{dx}(P) = \frac{3x_P^2 + a}{2y_P}$ respectively while adding or doubling on the curve, we clearly see that these computations involve not only a certain number of Multiplications (M) and Squarings (S), but also Inversions (I). Since computations are done in $\mathbb{Z}/n\mathbb{Z}$ with $n = p \times q$, p, q primes, denominators $x_P - x_Q$ and $2y_P$ of α are invertible if and only if $\gcd(x_P - x_Q, n) = 1$ when adding, or $\gcd(2y_P, n) = 1$ when doubling. If not equal to 1 and $\neq n$ then a proper divisor or a non-trivial factor of n is found. This is the all idea of ECM, using the isomorphism between $E(\mathbb{Z}/n\mathbb{Z})$ and $E(\mathbb{F}_p) \times E(\mathbb{F}_q)$.

Given a bound B , a point $P \in E(\mathbb{Z}/n\mathbb{Z})$ and a B -smooth scalar k , the method consists on computing kP until finding $kP = O$ using double and add method or NAF (see [58, 70]) to speed-up computations, (see [11–14] for improvements, and parametrizations yielding fewer number of M, S, and I) For performances purpose, large torsion subgroup elliptic curves over rationals present much more advantages. Some families and parametrizations of curves present better performance compared to the conventional short weierstrass equation in characteristic greater than 3. By Mazur theorem, every elliptic curve has a group torsion subgroup isomorphic to either one of $E_{tors}(\mathbb{Q}) \cong \mathbb{Z}/n_i\mathbb{Z}$, $n_i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$ or 12 or $E_{tors}(\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2n_j\mathbb{Z}$, $n_j = 1, 2, 3$ or 4. So curves whose points have large torsion subgroup over $\mathbb{Q}$ are preferable.

For example over a field $\mathbb{K}$, the Montgomery curve given by $M_{A,B} : By^2 = x^3 + Ax^2 + x$ where $B(A^2-4) \neq 0$ used in GMP-ECM implementation (see [59]) with the law $\forall P = (x_p, y_p), Q = (x_q, y_q) \in \{(x, y) \in \mathbb{K}^2 / By^2 = x^3 + Ax^2 + x\}$,

$$P+Q = \left(\frac{B(x_q y_p - x_p y_q)^2}{x_p x_q (x_q - x_p)^2}, \frac{(2x_p + x_q + A)(y_q - y_p)}{x_q - x_p} - \frac{B(y_q - y_p)^3}{(x_q - x_p)^3} - y_p \right) \text{ and}$$

equivalent to the Weierstrass reduced form through the map $\varphi : M_{A,B} \rightarrow E_{a,b}, (x, y) \mapsto (x', y') = (\frac{x}{B} + \frac{A}{3B}, \frac{y}{B})$ where $a = \frac{3-A^2}{3B^2}, b = \frac{2A^3-9A}{27B^3}$, and more recently Edwards Curves given by $x^2 + y^2 = 1 + dx^2 y^2$ with $d \setminus \{0, 1\}$ and generally twisted edwards curve given by $E_{a,d} : ax^2 + y^2 = 1 + dx^2 y^2$, with $a^5 \neq a, d \setminus \{0, 1\}$ used in the latest implementation ECM-MPFQ (see [30]) present better performances for ECM. The Edward's addition law is defined by $\forall P = (x_p, y_p), Q = (x_q, y_q) \in \{(x, y) \in \mathbb{K}^2 : ax^2 + y^2 = 1 + dx^2 y^2\}$,

$$P+Q = \left(\frac{1}{1+dx_p x_q y_p y_q} (x_p y_q + x_q y_p), \frac{1}{1-dx_p x_q y_p y_q} (y_p y_q - ax_p x_q) \right)$$

which is equivalent to

$$P+Q = \left(\frac{1}{ax_p x_q + y_p y_q} (x_p y_p + x_q y_q), \frac{1}{x_p y_q - y_p x_q} (x_p y_p - x_q y_q) \right)$$

(see [43]) where point doubling is not well defined and any Edward's equation form is birationally equivalent to the Montgomery form through the isomorphism $\Omega : E_{a,d} \rightarrow M_{A,B}, (x, y) \mapsto (x', y') = (\frac{1+y}{1-y}, \frac{1+y}{x(1-y)})$ where $A = \frac{2(a+d)}{a-d}, B = \frac{4}{a-d}$ and $\Omega^{-1} : M_{A,B} \rightarrow E_{a,d}, (x', y') \mapsto (\frac{x'}{y'}, \frac{x'-1}{x'+1})$ where $a = \frac{A+2}{B}, d = \frac{A-2}{B}$.

(5) Special number field sieve

The Special Number Field Sieve (SNFS) is one of the fastest known algorithms for factoring large integers of a special form, such as numbers of the form $a^n \pm b$. It uses algebraic number fields and a lattice reduction algorithm to find factors efficiently.

Its time complexity is expected to be

$$O\left(e^{(1+O(1))\left(\frac{32}{9}\log n\right)^{1/3}(\log \log n)^{2/3}}\right).$$

2.2 General purpose factorization methods

This category of factorization methods has a running time entirely depending on the size of the integer to be factored. Among, we have:

(1) Trial division

The trial division factorization method consists on dividing the number n by successive integers starting from the smallest prime (2) until the square root of n . If it is divisible by one of these integers, that integer is a factor of n .

Its worst case time complexity is $O(\sqrt{n})$.

(2) Dixon's factorization method

The Dixon's Factorization Method is a probabilistic algorithm that finds factors of a composite number n by searching for numbers x and y such that $x^2 \equiv y^2 \pmod{n}$. It uses a set of smooth numbers to find these congruences.

In the worst case, it has a time complexity of

$$O\left(e^{2\sqrt{2}\sqrt{\log n \log \log n}}\right) = L_n\left[\frac{1}{2}, 2\sqrt{2}\right].$$

(3) Continued fraction factorization method

The continued fraction factorization method (CFRAC) is a general purpose integer factorization algorithm, introduced

by D. H. Lehmer and R. E. Powers in 1931 and developed by Michael A. Morrison and John Brillhart in 1975.

This method is based on Dixon's factorization method. It uses convergents in the regular continued fraction expansion of $\sqrt{kn}$ for any positive integer k .

It has a complexity of $O\left(e^{\sqrt{2\log n \log \log n}}\right) = L_n\left[1/2, \sqrt{2}\right]$.

(4) Quadratic sieve

The Quadratic Sieve (QS) is a factorization algorithm that finds factors of a composite number n by searching for numbers x and y such that $x^2 \equiv y^2 \pmod{n}$. It uses a large set of smooth numbers to build a congruence of squares.

Its expected time complexity is $O\left(e^{(1+O(1))\sqrt{\ln n \ln \ln n}}\right)$.

(5) General number field sieve

The General Number Field Sieve (GNFS) is the most efficient classical algorithm for factoring large integers, particularly those with more than 100 digits. It uses algebraic number fields and polynomial selection to find relations that lead to a factorization of the target composite number n .

Its expected time complexity is

$$O\left(e^{\left(\left(\left(\frac{8}{3}\right)^{2/3} + O(1)\right)(\log n)^{1/3}(\log \log n)^{2/3}\right)}\right) = L_n\left[\frac{1}{3}, (64/9)^{1/3}\right].$$

Summary of some of the main factorization Methods:

In the same setting as above, we consider n a semiprime of size b bits, p and q its factors with $\min\{p, q\} = p$. We then have the following summary table

Table 1: Integer factorization Methods by category

Title	Special purpose	General Purpose	Quantum	Complexity
Trial Division	✗	✓	✗	$O(\sqrt{n})$
Pollard's Rho	✓	✗	✗	$O(\sqrt{p})$
Fermat's Method	✓	✗	✗	$O(n^{1/4})$
Euler's Method	✓	✗	✗	$O(\sqrt{p})$
Lenstra's Method	✓	✗	✗	$O\left(e^{(\sqrt{2}+O(1))\sqrt{\log p \log \log p}}\right)$
Special Number field sieve	✓	✗	✗	$O\left(e^{(1+O(1))\left(\frac{32}{9}\log n\right)^{1/3}(\log \log n)^{2/3}}\right)$
Shor's algorithm	✗	✓	✓	$O(b^3)$
Dixon's Method	✗	✓	✗	$O\left(e^{2\sqrt{2}\sqrt{\log n \log \log n}}\right)$
Quadratic Sieve	✗	✓	✗	$O\left(e^{(1+O(1))\sqrt{\ln n \ln \ln n}}\right)$
General Number Field Sieve	✗	✓	✗	$O\left(e^{\left(\left(\left(\frac{8}{3}\right)^{2/3} + O(1)\right)(\log n)^{1/3}(\log \log n)^{2/3}\right)}\right)$

* Table of integer factorization methods by category.

3 CONTRIBUTIONS

The most difficult case in integer factorization is to factor a semi prime whose factors are big primes.

For this reason, in this section we consider the modulus n to be a semiprime.

3.1 Rectangle point of view (Rpv)

First and foremost, integer factorization can be reformulated as follows: the factorization of a semiprime $n = pq$ is equivalent to finding a rectangle's length and width whose area is known as n . Equivalently, it is to find a perimeter of a rectangle whose area is known.

This is proved by the fact this case the perimeter $p = 2(p + q)$ and area $n = pq$.

Then recovering p and q becomes equivalent to either finding bounds p and q such that

$$\int_0^p \int_0^q d_x d_y = n$$

, either finding primitives $\gamma_1(x)$ and $\gamma_2(y)$ such that for given bounds $(\alpha_0, \alpha_1, \beta_0, \beta_1) \in \mathbb{Z}^4$, we have

$$\int_{\alpha_0}^{\alpha_1} \int_{\beta_0}^{\beta_1} \gamma_1(x) \gamma_2(y) d_x d_y = n$$

.

LEMMA 3.1.

For any prime number p greater than 3, there exists a positive integer x such that $p = y(x)$ is solution of the first order ordinary differential equation with Dirichlet condition

$$y' = 6, \text{ with } y(0) = \pm 1$$

PROOF.

$$y' = \frac{dy}{dx} = 6,$$

then $\int dy = \int 6dx$. Thus $y = 6x + c$ with $c = \pm 1$. Hence the prime number $p = y(x) = 6x \pm 1$. From a basic observation from elementary number theory using properties of integers and modular arithmetic, any prime number can be represented as such. $\square$

Example 3.2.

Consider the following primes:

- the 100 bits prime:

$$p = 570303428823043591555786898897 = y(x) = 6x - 1, \\ \text{with } x = 95050571470507265259297816483$$

- the 233 bits prime:

$$p = 6833702715893496540959299291382359232559264092523368475576074466071807 = y(x) = 6x - 1 \text{ with } x = 1138950452648916090159883215230393205426544015420561412596012411011968$$

- the 333 bits prime:

$$p = 6358337399459401417678277930704848782199223004051687909162273047126596686480950 \\ 628594226066122559163 = y(x) = 6x + 1 \text{ with } x = 105972289909900236279712988450808130 \\ 366537167341947984860378841187766114413491771432371011020426527$$

- the 1000 bits prime:

$$p = 658089189840301730450496429519238792660987613697354796644846033244621236257652046 \\ 02174048492175950412255940691871306258635428762907108947956854239712936130938554245 \\ 542843969795828244216429256228793034041405027874070693186794739605844025450561774421 \\ 858331525104830589953173100274244230949509794254643 = y(x) = 6x - 1 \text{ with } x = \\ 109681531640050288408416071586539798776831268949559132774141005540770206042942007670 \\ 290080820293250687092656781978551043105904793817851491326142373285489355156425707590 \\ 473994965971374036071542704798839006900837979011782197799123267640670908426962403643 \\ 055254184138431658862183379040705158251632375774$$

Let's consider the following proposition:

PROPOSITION 3.3.

Given a RSA modulus $n \geq 35$, set $\gamma_1(x) = 6$; $\gamma_2(y) = 6$ and $\alpha_0 = \beta_0 = \pm \frac{1}{6}$ then there exists α and β such that

$$\int_{\pm \frac{1}{6}}^{\alpha} \int_{\pm \frac{1}{6}}^{\beta} \gamma_1(x) \gamma_2(y) d_x d_y = n$$

.

PROOF.

Using the Fubini-Tonelli [8] theorem,

$$\begin{aligned} \int_{\pm \frac{1}{6}}^{\alpha} \int_{\pm \frac{1}{6}}^{\beta} \gamma_1(x) \gamma_2(y) d_x d_y &= \int_{\pm \frac{1}{6}}^{\alpha} \gamma_1(x) d_x \int_{\pm \frac{1}{6}}^{\beta} \gamma_2(y) d_y \\ &= \int_{\pm \frac{1}{6}}^{\alpha} (6) d_x \int_{\pm \frac{1}{6}}^{\beta} (6) d_y \\ &= [6x + C_1]_{\pm \frac{1}{6}}^{\alpha} [6y + C_2]_{\pm \frac{1}{6}}^{\beta} \\ &= (6\alpha \pm 1) (6\beta \pm 1) \\ &= y(\alpha) y(\beta), \text{ from lemma 3.1} \\ &= pq \\ &= n \end{aligned}$$

$\square$

Example 3.4.

Here we give two examples of 256 bits semiprimes.

- $\alpha = 31184864858157931962611989461653189383$;
 $\beta = 56155612513944961522154793570341370125$

$$\begin{aligned} \int_{-\frac{1}{6}}^{\alpha} \int_{-\frac{1}{6}}^{\beta} \gamma_1(x) \gamma_2(y) d_x d_y &= \int_{-\frac{1}{6}}^{\alpha} \gamma_1(x) d_x \int_{-\frac{1}{6}}^{\beta} \gamma_2(y) d_y \\ &= \int_{-\frac{1}{6}}^{31184864858157931962611989461653189383} (6) d_x \\ &\times \int_{-\frac{1}{6}}^{56155612513944961522154793570341370125} (6) d_y \\ &= [6x + C_1]_{-\frac{1}{6}}^{31184864858157931962611989461653189383} \\ &\times [6y + C_2]_{-\frac{1}{6}}^{56155612513944961522154793570341370125} \\ &= (6 \cdot 31184864858157931962611989461653189383 + 1) \\ &\times (6 \cdot 56155612513944961522154793570341370125 + 1) \\ &= y(31184864858157931962611989461653189383) \\ &\times y(56155612513944961522154793570341370125) \\ &= 187109189148947591775671936769919136299 \\ &\times 336933675083669769132928761422048220751 \\ &= 6304338674188041713811886561148660301051 \\ &9216638181159430642741072139609140549 \\ &= n \end{aligned}$$

, from lemma 3.1

- $\alpha = 42575376056348197982869887629770563187$;
 $\beta = 33263421842796264531163131543015665831$

$$\begin{aligned}
 \int_{\frac{1}{6}}^{\alpha} \int_{-\frac{1}{6}}^{\beta} \gamma_1(x) \gamma_2(y) dx dy &= \int_{\frac{1}{6}}^{\alpha} \gamma_1(x) dx \int_{-\frac{1}{6}}^{\beta} \gamma_2(y) dy \\
 &= \int_{\frac{1}{6}}^{42575376056348197982869887629770563187} (6) dx \\
 &\times \int_{-\frac{1}{6}}^{33263421842796264531163131543015665831} (6) dy \\
 &= [6x + C_1]_{\frac{1}{6}}^{42575376056348197982869887629770563187} \\
 &\times [6y + C_2]_{-\frac{1}{6}}^{33263421842796264531163131543015665831} \\
 &= (6 \cdot 42575376056348197982869887629770563187 - 1) \\
 &\times (6 \cdot 33263421842796264531163131543015665831 + 1) \\
 &= y(42575376056348197982869887629770563187) \\
 &\times y(33263421842796264531163131543015665831) \\
 &, \text{ from lemma 3.1} \\
 &= 255452256338089187897219325778623379121 \\
 &\times 199580531056777587186978789258093994987 \\
 &= 509832969796079184022321203452975440752167 \\
 &80739063868280880006751616374466427 \\
 &= n
 \end{aligned}$$

Additionally, in the same perspective, we have the following result, a new vulnerability on a RSA type modulus:

THEOREM 3.5.

Let n be a RSA modulus. If the diophantine equation $y^2 - 2^3n - 1 = 0$ has any solution in y , then RSA is broken in constant time complexity $O(1)$ in best case or in logarithmic time complexity $O(\log(8n+1))$ in the worst case.

PROOF.

Let $n = pq$ a semiprime with $q < p$.

Assume $n = \sum_{l=1}^p l = 1 + 2 + 3 + \dots + p$.

From the Gauss sum sequence, $n = S_p = p \frac{p+1}{2}$ or $n = \binom{n+1}{2}$ in binomial coefficient representation. This represents the area of a rectangle whose width is p and height is $\frac{p+1}{2}$. In other words, p verifies the equations

$$y^2 + y - 2n = 0 \quad (1)$$

and

$$x^2 - \left(\frac{3p+1}{2}\right)x + n = 0 \quad (2)$$

from $x^2 - Sx + P = 0$ result.

Considering equation (1), this equation yields integer roots if and only if its discriminant $1 + 2^3n$ is a square. Equivalently, there exists an integer y such that $y^2 - 2^3n - 1 = 0$. In this case $p = \frac{|-1 \pm (1 + 2^3n)^{1/2}|}{2}$. Finally, depending on the square root computation algorithm, its complexity is at best in $O(1)$ or at worst in $O(\log(8n+1))$. $\square$

Remark

This theorem can be reformulated as follows:

If a RSA modulus is a triangular number, then RSA is broken in at least constant time complexity and at most logarithmic time complexity.

Example 3.6.

- $n = 15$, $8n + 1 = 121$ is a square, then $p = |5| = 5$ is a prime factor of n : $n = 3 \times 5$.
- $n = 25651$, $8n + 1 = 205209$ is a square, then $p = |-227| = 227$ is a prime factor of n : $n = 113 \times 227$.
- $n = 16744225501$, $8n + 1 = 133953804009$ is a square, then $p = |-182999| = 182999$.
- $n = 22008842474653$, $8n + 1 = 176070739797225$ is a square, then $p = |-6634583| = 6634583$.
- $n = 21997001338003$, $8n + 1 =$ is a square, then $p = |6632797| = 6632797$.

3.2 Matrix Decomposition point of view (MDpv)

Here we present another approach, consisting in considering the modulus n as a matrix determinant.

THEOREM 3.7.

Let $n = pq$ be a RSA modulus, then there exists a matrix decomposition N of n of size 2 such that $N = PQ$ with $\det N = n$, $\det P = p$ and $\det Q = q$.

PROOF.

Triivially, one can find non zero integers a and b so that applying Bezout's theorem, there exists integers d, c such that $ad - bc = \gcd(a, b) = n$. Let $N = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $P = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix}$ and $Q = \begin{pmatrix} y_1 & y_2 \\ y_3 & y_4 \end{pmatrix}$ $\in M_2(\mathbb{Z})$. $n = pq$ if and only if

$ad - bc = (x_1x_4 - x_2x_3)(y_1y_4 - y_2y_3)$ or equivalently $\det N = \det P \cdot \det Q$

$$PQ = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} \begin{pmatrix} y_1 & y_2 \\ y_3 & y_4 \end{pmatrix} = \begin{pmatrix} x_1y_1+x_2y_3 & x_1y_2+x_2y_4 \\ x_3y_1+x_4y_3 & x_3y_2+x_4y_4 \end{pmatrix} \text{ Then}$$

$$x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4 \text{ satisfy } \begin{cases} x_1y_1+x_2y_3 = a \\ x_1y_2+x_2y_4 = b \\ x_3y_1+x_4y_3 = c \\ x_3y_2+x_4y_4 = d \end{cases} \quad (2)$$

From this system,

$$x_1 = \frac{ay_4 - by_3}{y_1y_4 - y_2y_3}, x_2 = \frac{ay_2 - by_1}{y_1y_4 - y_2y_3}, x_3 = \frac{cy_4 - dy_3}{y_1y_4 - y_2y_3}, x_4 = \frac{cy_2 - dy_1}{y_1y_4 - y_2y_3} \text{ We verify that:}$$

$$\begin{aligned}
 x_1x_4 - x_2x_3 &= \frac{ay_4 - by_3}{y_1y_4 - y_2y_3} \times \frac{dy_1 - cy_2}{y_1y_4 - y_2y_3} - \frac{cy_4 - dy_3}{y_1y_4 - y_2y_3} \times \frac{by_1 - ay_2}{y_1y_4 - y_2y_3} \\
 &= \frac{(ay_4 - by_3)(dy_1 - cy_2) - (cy_4 - dy_3)(by_1 - ay_2)}{(y_1y_4 - y_2y_3)^2}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{(bc - ad) y_2 y_3 + (ad - bc) y_1 y_4 + (ac - ac) y_2 y_4 + (bd - bd) y_1 y_3}{(y_1 y_4 - y_2 y_3)^2} \\
&= \frac{(bc - ad) y_2 y_3 + (ad - bc) y_1 y_4}{(y_1 y_4 - y_2 y_3)^2} \\
&= \frac{(ad - bc) (y_1 y_4 - y_2 y_3)}{(y_1 y_4 - y_2 y_3)^2} \\
&= \frac{ad - bc}{y_1 y_4 - y_2 y_3} \\
&= \frac{n}{y_1 y_4 - y_2 y_3} \\
&= x_1 x_4 - x_2 x_3
\end{aligned}$$

□

From this, we have:

- $\gcd(\det P, n) \mid n$,
then $\gcd(x_1 x_4 - x_2 x_3, n) \mid n$, hence $\gcd(x_1 x_4 - x_2 x_3, n) = y_1 y_4 - y_2 y_3$
- $\gcd(\det Q, n) \mid n$,
then $\gcd(y_1 y_4 - y_2 y_3, n) \mid n$, hence $\gcd(y_1 y_4 - y_2 y_3, n) = x_1 x_4 - x_2 x_3$

Hence we have the following new system from (2).

$$\begin{cases} x_1 y_1 + x_2 y_3 = a \\ x_1 y_2 + x_2 y_4 = b \\ x_3 y_1 + x_4 y_3 = c \\ x_3 y_2 + x_4 y_4 = d \\ (x_1 x_4 - x_2 x_3) (y_1 y_4 - y_2 y_3) = n \end{cases} \quad (3)$$

Which is a non linear system of 5 equations with 8 unknowns.

System resolution

In this subsection, we propose some approaches to solve this system.

Gröbner Basis

We apply gröbner basis to the resolution of this system of equations. We start by setting the polynomials to zero as follows:

$$\begin{cases} x_1 y_1 + x_2 y_3 - a = 0 \\ x_1 y_2 + x_2 y_4 - b = 0 \\ x_3 y_1 + x_4 y_3 - c = 0 \\ x_3 y_2 + x_4 y_4 - d = 0 \\ (x_1 x_4 - x_2 x_3) (y_1 y_4 - y_2 y_3) - n = 0 \end{cases} \quad (4)$$

Since $x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4$ are integers, we consider the polynomial ring $\mathbb{Z}[x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4]$.

Another consideration is the lexicographic order as monomial order given by

$$x_1 > x_2 > x_3 > x_4 > y_1 > y_2 > y_3 > y_4$$

and finally we consider the ideal I generated by f_1, f_2, f_3, f_4 and f_5 as follows $I = \langle f_1, f_2, f_3, f_4, f_5 \rangle$ over $\mathbb{Z}[x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4]$ where $f_1 = x_1 y_1 + x_2 y_3 - a$;

$f_2 = x_1 y_2 + x_2 y_4 - b$; $f_3 = x_3 y_1 + x_4 y_3 - c$;

$f_4 = x_3 y_2 + x_4 y_4 - d$; and $f_5 = (x_1 x_4 - x_2 x_3) (y_1 y_4 - y_2 y_3) - n$.

We then apply the Buchberger's algorithm to generate the gröbner basis.

Let G be the gröbner basis, we initialize G as $G = \{f_1, f_2, f_3, f_4, f_5\}$ and we compute the S-polynomials as follows: for each pair of polynomials f_i and f_j in G ,

$$S(f_i, f_j) = \frac{Lcm(Lm(f_i), Lm(f_j))}{Lt(f_i)} f_i - \frac{Lcm(Lm(f_i), Lm(f_j))}{Lt(f_j)} f_j$$

where Lcm is the least common multiple, Lt the leading term and Lm the leading monom.

Since our ideal I is generated by 5 polynomials, we expect the set S of S-polynomials to have a cardinal of $Card(S) = \binom{5}{2} = \frac{5!}{2!3!} = 10$.

Computation of S-Polynomials:

$$Lt(f_1) = Lm(f_1) = x_1 y_1; \quad Lt(f_2) = Lm(f_2) = x_1 y_2;$$

$$Lt(f_3) = Lm(f_3) = x_3 y_1; \quad Lt(f_4) = Lm(f_4) = x_3 y_2;$$

$$Lt(f_5) = Lm(f_5) = x_1 x_4 y_1 y_4.$$

We get:

$$S(f_1, f_2) = y_2 f_1 - y_1 f_2; \quad S(f_1, f_3) = x_3 f_1 - x_1 f_3;$$

$$S(f_1, f_4) = x_3 y_2 f_1 - x_1 y_1 f_4; \quad S(f_1, f_5) = x_4 y_4 f_1 - f_5;$$

$$S(f_2, f_3) = x_3 y_1 f_2 - x_1 y_2 f_3; \quad S(f_2, f_4) = x_3 y_1 f_2 - x_1 y_1 f_4;$$

$$S(f_2, f_5) = x_4 y_1 y_4 f_2 - y_2 f_5; \quad S(f_3, f_4) = y_2 f_3 - y_1 f_4;$$

$$S(f_3, f_5) = x_1 x_4 y_4 f_3 - x_3 f_5; \quad S(f_4, f_5) = x_1 x_4 y_1 y_4 f_4 - x_3 y_2 f_5;$$

In the polynomial ring $\mathbb{Z}[x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4]$, a Gröbner basis is:

$$ax_3 - cx_1 + x_1 x_4 y_3 - x_2 x_3 y_3 = 0 \quad (1),$$

$$bx_3 - dx_1 + x_1 x_4 y_4 - x_2 x_3 y_4 = 0 \quad (2),$$

$$-a + x_1 y_1 + x_2 y_3 = 0, -b + x_1 y_2 + x_2 y_4 = 0 \quad (3),$$

$$a^2 x_3 y_4 - abx_3 y_3 - acx_1 y_4 + bcx_1 y_3 + nx_1 y_3 = 0 \quad (4),$$

$$ax_3 y_4 - bx_3 y_3 - cx_1 y_4 + dx_1 y_3 = 0 \quad (5),$$

$$ay_2 - by_1 + x_2 y_1 y_4 - x_2 y_2 y_3 = 0 \quad (6),$$

$$a^2 x_4 y_2 - abx_4 y_1 - acx_2 y_2 + bcx_2 y_1 + nx_2 y_1 = 0 \quad (7),$$

$$ax_4 y_2 - bx_4 y_1 - cx_2 y_2 + dx_2 y_1 = 0 \quad (8),$$

$$a^2 x_4 y_4 - abx_4 y_3 - acx_2 y_4 - an + bcx_2 y_3 + nx_2 y_3 = 0 \quad (9),$$

$$ax_4 y_4 - bx_4 y_3 - cx_2 y_4 + dx_2 y_3 - n = 0 \quad (10),$$

$$-c + x_3 y_1 + x_4 y_3 = 0, -d + x_3 y_2 + x_4 y_4 = 0 \quad (11),$$

$$cy_2 - dy_1 + x_4 y_1 y_4 - x_4 y_2 y_3 = 0 \quad (12),$$

$$ad - bc - n = 0 \quad (13)$$

given in term of a system of equations as

$$\begin{cases} \begin{cases} ad - bc - n = 0 \\ x_3y_1 + x_4y_3 - c = 0 \\ x_1y_1 + x_2y_3 - a = 0 \\ x_1y_2 + x_2y_4 - b = 0 \\ x_3y_2 + x_4y_4 - d = 0 \end{cases} \\ cy_2 - dy_1 + x_4y_1y_4 - x_4y_2y_3 = 0 \\ ax_3 - cx_1 + x_1x_4y_3 - x_2x_3y_3 = 0 \\ ay_2 - by_1 + x_2y_1y_4 - x_2y_2y_3 = 0 \\ bx_3 - dx_1 + x_1x_4y_4 - x_2x_3y_4 = 0 \\ ax_3y_4 - bx_3y_3 - cx_1y_4 + dx_1y_3 = 0 \\ ax_4y_2 - bx_4y_1 - cx_2y_2 + dx_2y_1 = 0 \\ ax_4y_4 - bx_4y_3 - cx_2y_4 + dx_2y_3 - n = 0 \\ a^2x_3y_4 - abx_3y_3 - acx_1y_4 + bcx_1y_3 + nx_1y_3 = 0 \\ a^2x_4y_2 - abx_4y_1 - acx_2y_2 + bcx_2y_1 + nx_2y_1 = 0 \\ a^2x_4y_4 - abx_4y_3 - acx_2y_4 - an + bcx_2y_3 + nx_2y_3 = 0 \end{cases} \quad (5)$$

Discussion:

A part from trying to find factors p and q through system resolution, one can for different matrix decompositions N of n , diagonalize N if $(a+d)^2 - 4n$ is a square. Indeed the characteristic polynomial in this case $P(\lambda) = (a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + n$, which is diagonalizable over $\mathbb{Z}$ only if the discriminant of the characteristic polynomial is a square.

Example 3.8.

Let $n = 98297604265312920805982784738341591888656111416198453609045170174910824769389$ with $a = 98297604265312920805982784738341591889580784184957701040844308085531017447603$;
 $b = 98297604265312920805982784738341591890174115698218506161959832718260667360288$
 $c = 331341255498442310683613277890542765532$

$$d = 331341255498442310683613277890542765535 \quad N = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Considering the System 5, a solution of the system gives: $x_1 = 296665756630402560557762316364824956342$;

$x_2 = 296665756630402560557762316364824956343$;

$x_3 = 1$; $x_4 = 2$.

$y_1 = 331341255498442310683613277890542765530$;

$y_2 = 331341255498442310683613277890542765531$;

$y_3 = 1$; $y_4 = 2$; Hence $P = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix}$ and $Q = \begin{pmatrix} y_1 & y_2 \\ y_3 & y_4 \end{pmatrix} \in M_2(\mathbb{Z})$,

$p = \det(P) = x_1x_4 - x_2x_3 = 296665756630402560557762316364824956341$ and $q = \det(Q) = y_1y_4 - y_2y_3 = 331341255498442310683613277890542765529$

3.3 Modulus Algebraic Form Point of View (MAFpv)

In this subsection, we present another approach that consists in considering different algebraic forms of the semiprime factors.

THEOREM 3.9.

Any RSA modulus, particularly any semiprime n can be represented in the ring $\mathbb{Z}[x, y]$ as one of the following:

$$n(x, y) = \begin{cases} 36xy - 6(x+y) + 1 \\ 36xy + 6(x+y) + 1 \\ 36xy + 6(x-y) - 1 \\ 36xy - 6(x-y) - 1 \end{cases}$$

PROOF.

From Lemma 3.1, there exist differential equations $t' = 6$ with $t(0) = \pm 1$ and $z' = 6$ with $z(0) = \pm 1$ such that $n(x, y) = z(x)t(y)$. We then distinguish the following cases:

If $t(0) = -1$ and $z(0) = -1$: $t(y) = 6y - 1$ and $z(x) = 6x - 1$. Then $n(x, y) = z(x)t(y) = (6x - 1)(6y - 1) = 36xy - 6(x+y) + 1$.

If $t(0) = 1$ and $z(0) = 1$: $t(y) = 6y + 1$ and $z(x) = 6x + 1$. Then $n(x, y) = z(x)t(y) = (6x + 1)(6y + 1) = 36xy + 6(x+y) + 1$.

If $t(0) = -1$ and $z(0) = 1$: $t(y) = 6y - 1$ and $z(x) = 6x + 1$. Then $n(x, y) = z(x)t(y) = (6x + 1)(6y - 1) = 36xy - 6(x-y) - 1$.

If $t(0) = 1$ and $z(0) = -1$: $t(y) = 6y + 1$ and $z(x) = 6x - 1$. Then $n(x, y) = z(x)t(y) = (6x - 1)(6y + 1) = 36xy + 6(x-y) - 1$.

All these can be re-written as:

$$n(x, y) = \begin{cases} 36xy - 6(x+y) + 1 \\ 36xy + 6(x+y) + 1 \\ 36xy + 6(x-y) - 1 \\ 36xy - 6(x-y) - 1 \end{cases} \quad \text{or } n(x, y) = \begin{cases} 36xy + 6|x+y| + 1 \\ 36xy + 6|x-y| - 1 \end{cases}$$

□

REMARK 3.1.

(1) The polynomials $n(x, y) \in \mathbb{Z}[x, y]$ as defined in the previous theorem have a total degree of $d = 2$, ie every monomial $x^i y^j$ satisfies $i + j \leq d = 2$

(2) We consider $X \in \mathbb{Z}_{>0}$ respectively $Y \in \mathbb{Z}_{>0}$, the upper bounds for x respectively for y and M a RSA type modulus greater than n .

From this remark, we have the following theorems:

THEOREM 3.10.

Let $\deg_x(n(x, y)), \deg_y(n(x, y)) < d = 2$;

$$n(x, y) = \sum_{0 \leq i, j < d} n_{i,j} \delta_{ij} x^i y^j = \begin{cases} 36xy + 6|x+y| + 1 \\ 36xy + 6|x-y| - 1 \end{cases}$$

where $n_{00} = 1, n_{11} = 36, n_{10} = n_{01} = 6$ and

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{else} \end{cases}$$

and for $|x| < X$ and $|y| < Y$, define $W = \max_{0 \leq i, j \leq d} |n_{i,j}| X^i Y^j$.

If $XY < W^{1/3} = 6^{2/3} X^{1/3} Y^{1/3}$ then we can recover solutions (x, y) in polynomial time in $\log(W)$ and 2^2 .

PROOF.

This is a direct application of coppersmith's theorem on bivariate polynomials [10]. □

From this first underlying theorem, the condition to recover solutions (x, y) in polynomial time cannot be achieved in practice. Hence, we consider the same theorem in a residual ring $\mathbb{Z}_M$ with M a RSA type modulus greater than n .

THEOREM 3.11.

Consider the polynomial $n(x, y) \in \mathbb{Z}[x, y]$ as defined above. Let M be a RSA type modulus greater than n , X and Y such that for all (x, y) solutions of $n(x, y)$, $|x| < X$ and $|y| < Y$. If $XY < M^{1/2-\epsilon}$ for some $0 < \epsilon < 1/2$ then we can compute $n(x, y) \equiv 0 \pmod{M}$ and find solutions of $n(x, y)$ over $\mathbb{Z}$.

PROOF. As with the previous theorem, this is a direct application of coppersmith's theorem on bivariate polynomials [10] on a residual ring. $\square$

4 CONCLUSION

In this paper, we have recalled some factorization methods and then approached the problem from completely different angles. First, we have considered the problem from the ring $(\mathbb{Z}, +, \cdot)$ to Lebesgue space $\mathcal{L}^1(X)$ where X can be $\mathbb{Q}$ or any given interval setting, in which case, this problem is equivalent to finding the perimeter of a rectangle whose area is known. In this direction, it is equivalent to either finding bounds of integrals or finding primitives for some given bounds. Secondly, we have taken the problem from the ring $(\mathbb{Z}, +, \cdot)$ to $(M_n(\mathbb{Z}), + \cdot)$ and show that this problem is equivalent to matrix decomposition, and therefore present a computing algorithm through a Gröbner basis computation approach and matrix diagonalization. Finally, we have addressed the problem depending on algebraic forms of factors and show that this problem is equivalent to finding small roots of a bivariate polynomial through coppersmith's method. The aim of this article is to present the problem from different angles, opening up new directions for approaching the problem in a different way taking advantage of measure, calculus and related theories.

As future perspective, for the MAFpv, a future immediate work would be investigating geometrically and finding a possible construction rule on these hyperbolas to recover the sought solutions with respect to the equation coefficients and divisors of $n - 1$. In addition, best upper bounds X and Y , and M should be investigated for better performances.

5 ACKNOWLEDGEMENTS

The authors would like to thank the anonymous reviewers for their helpful comments on the paper.

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